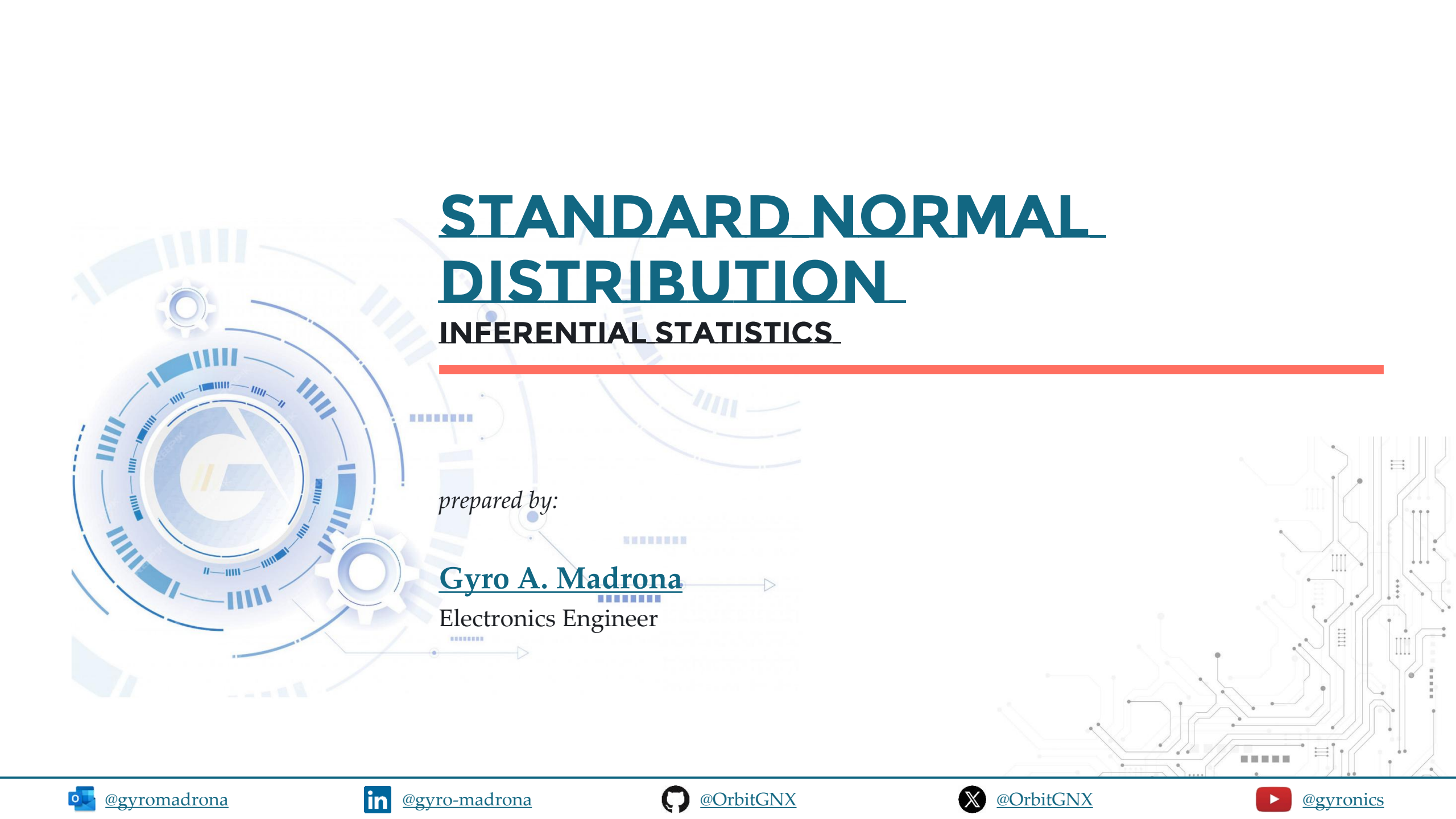




STANDARD NORMAL **DISTRIBUTION**

INFERENTIAL STATISTICS

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TOPIC OUTLINE

Standard Normal Distribution

Central Limit Theorem



STANDARD NORMAL DISTRIBUTION



STANDARDIZATION

Standardization is the process of converting the distribution of a variable $X \sim (\mu, \sigma^2)$ to a normal distribution $Z \sim N(0, 1)$.

Formula

$$Z = \frac{X - \mu}{\sigma}$$

(Note: An arrow points from x_i to the X in the formula.)

where

Z = z-score

X = random variable

example

Data	$X - \mu$	Z-score
1	$1-5 = -4$	-1.46
2	$2-5 = -3$	-1.095
3	$3-5 = -2$	-0.73 ← $\frac{3-5}{2.74}$
4	$4-5 = -1$	-0.365
5	$5-5 = 0$	0
6	$6-5 = 1$	0.365
7	$7-5 = 2$	0.73
8	$8-5 = 3$	1.095
9	$9-5 = 4$	1.46 ← $\frac{9-5}{2.74}$

$$\bar{X} = 5.0$$

$$S = 2.74$$



STANDARDIZATION

Standardization is the process of converting the distribution of a variable $X \sim (\mu, \sigma^2)$ to a normal distribution $Z \sim N(0, 1)$.

Formula

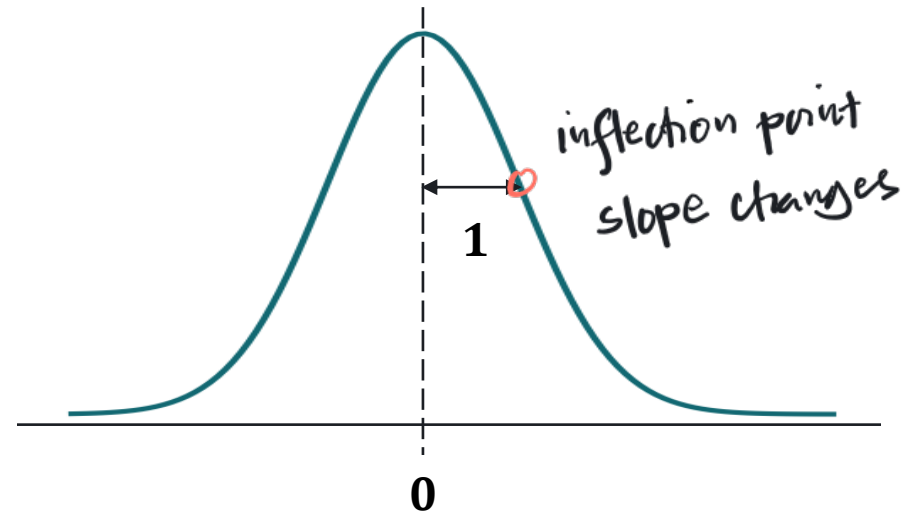
$$Z = \frac{X - \mu}{\sigma}$$

where

Z = z-score

X = random variable

Normal Distribution



STANDARD NORMAL DISTRIBUTION

Standard Normal Distribution

When we standardize the normal distribution $X \sim N(\mu, \sigma^2)$, the result is a standard normal distribution $Z \sim N(0, 1)$.

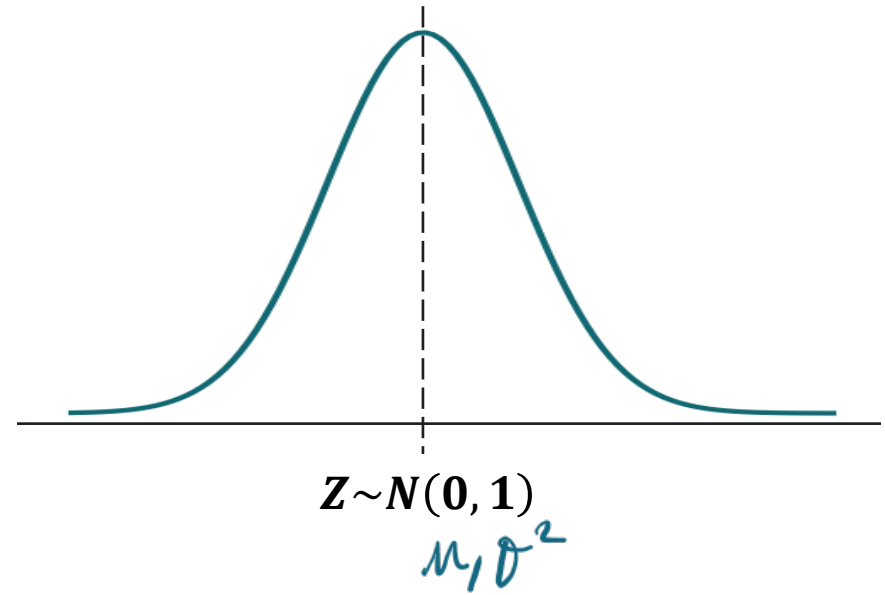
Formula

$$Z = \frac{X - \mu}{\sigma}$$

where

Z = z-score

X = random variable



EXERCISE

Convert the given dataset into a standard normal distribution $N(0, 1)$ by computing the **z-score** for each data point.

Data

1
2
2
3
3
3
4
4
5

$$\bar{X} = 3$$

Solution

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$$

$$\bar{X} = \frac{1 + 2 + 2 + \dots + 5}{9}$$

$$\bar{X} = 3$$



EXERCISE

Convert the given dataset into a standard normal distribution $N(0, 1)$ by computing the **z-score** for each data point.

Data

1
2
2
3
3
3
4
4
5

$$\bar{X} = 3 \quad S = 1.22$$

Solution

$$S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$$

$$S^2 = \frac{(1-3)^2 + (2-3)^2 + \dots + (5-3)^2}{9-1}$$

$$S^2 = 1.6$$

$$\underline{S = 1.22}$$



EXERCISE

Convert the given dataset into a standard normal distribution $N(0, 1)$ by computing the **z-score** for each data point.

Data	$X - \mu$	$z \leftarrow \frac{X - \mu}{s}$
1	-2	-1.64
2	-1	-0.82
2	-1	-0.82
3	0	0
3	0	0
3	0	0
4	1	0.82
4	1	0.82
5	2	1.64

$$\bar{X} = 3 \quad s = 1.22$$

Solution

Mean of z

$$\bar{z} = \frac{\sum_{i=1}^n z_i}{n}$$

$$\bar{z} = \frac{-1.64 - 0.82 + \dots + 1.64}{9}$$

$$\bar{z} = 0$$



EXERCISE

Convert the given dataset into a standard normal distribution $N(0, 1)$ by computing the **z-score** for each data point.

Data		$X - \mu$	$z \leftarrow \frac{X - \mu}{s}$
1	1-3	-2	-1.64
2	2-3	-1	-0.82
2	2-3	-1	-0.82
3	3-3	0	0
3	3-3	0	0
3	3-3	0	0
4	4-3	1	0.82
4	4-3	1	0.82
5	5-3	2	1.64

$$\bar{X} = 3 \quad s = 1.22$$

Solution

Variance of z

$$z^2 = \frac{\sum_{i=1}^n (z_i - \bar{z})^2}{n-1}$$

$$z^2 = \frac{(-1.64 - 0)^2 + (-0.82 - 0)^2 + \dots}{9-1}$$

$$\underline{z^2 = 1}$$

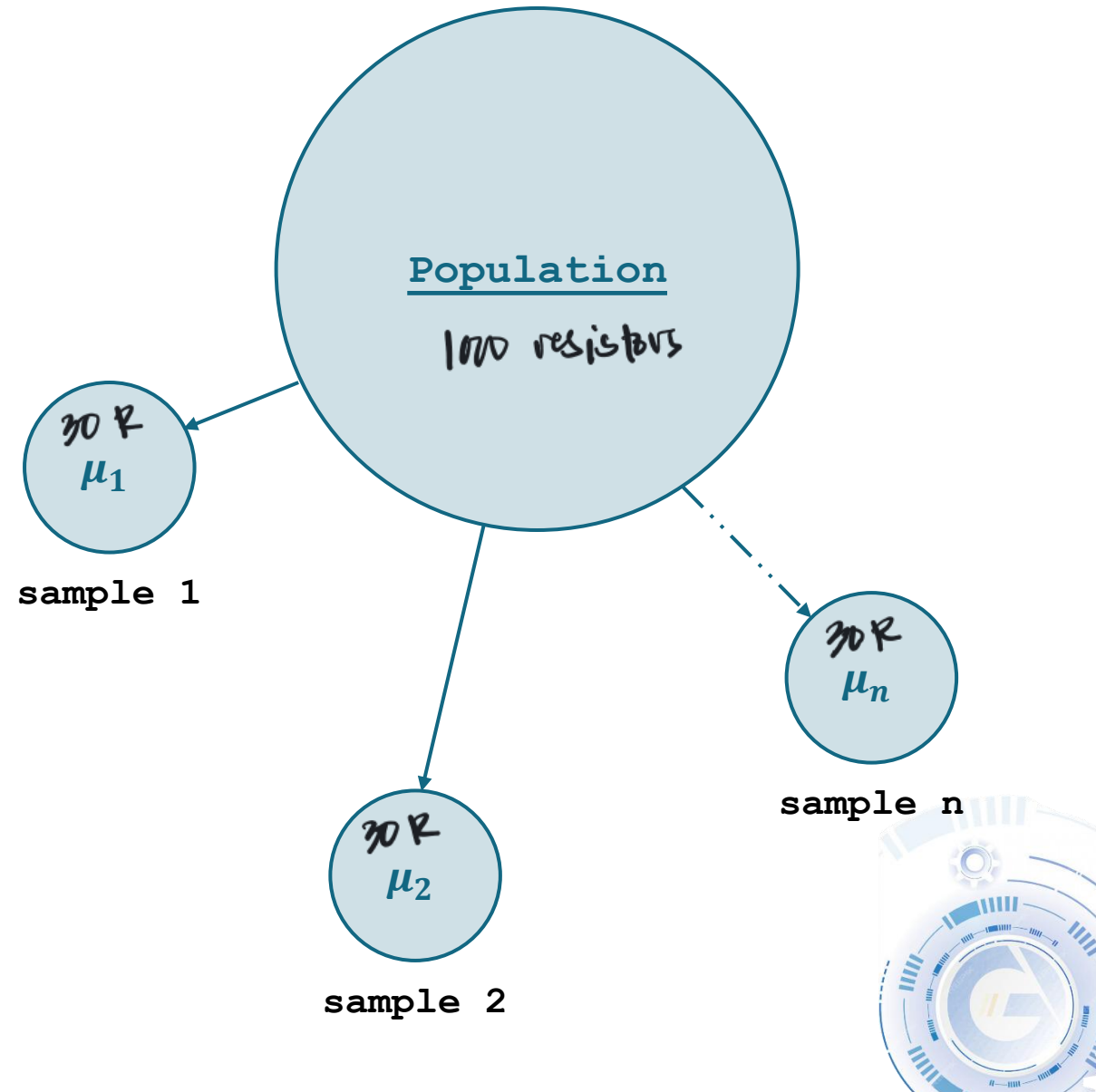


CENTRAL LIMIT THEOREM



CENTRAL LIMIT THEOREM

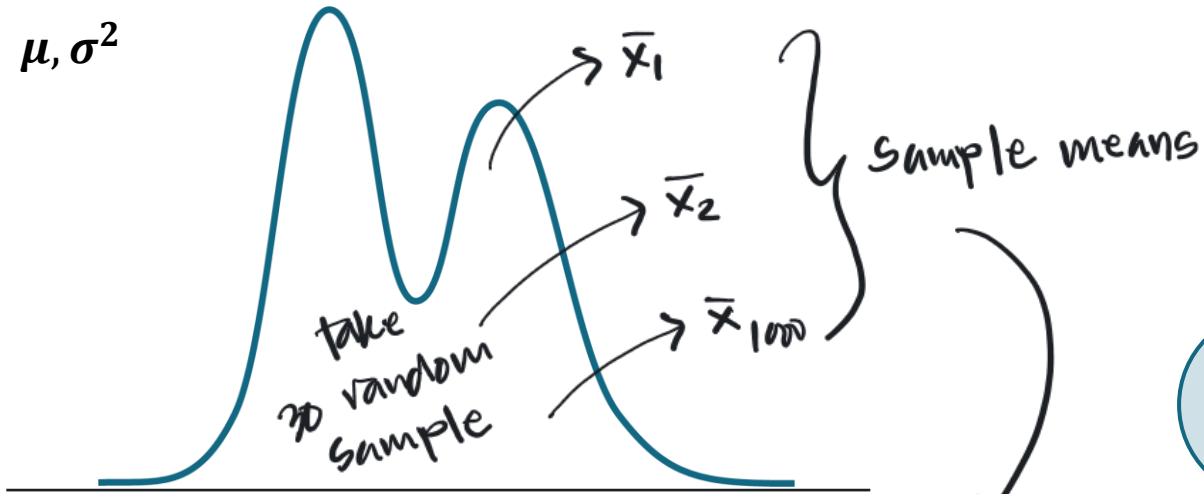
The Central Limit Theorem (CLT) states that the sampling distribution of the sample mean will be normally distributed, regardless of the shape of the original population distribution.



CENTRAL LIMIT THEOREM

Original Population Distribution

μ, σ^2



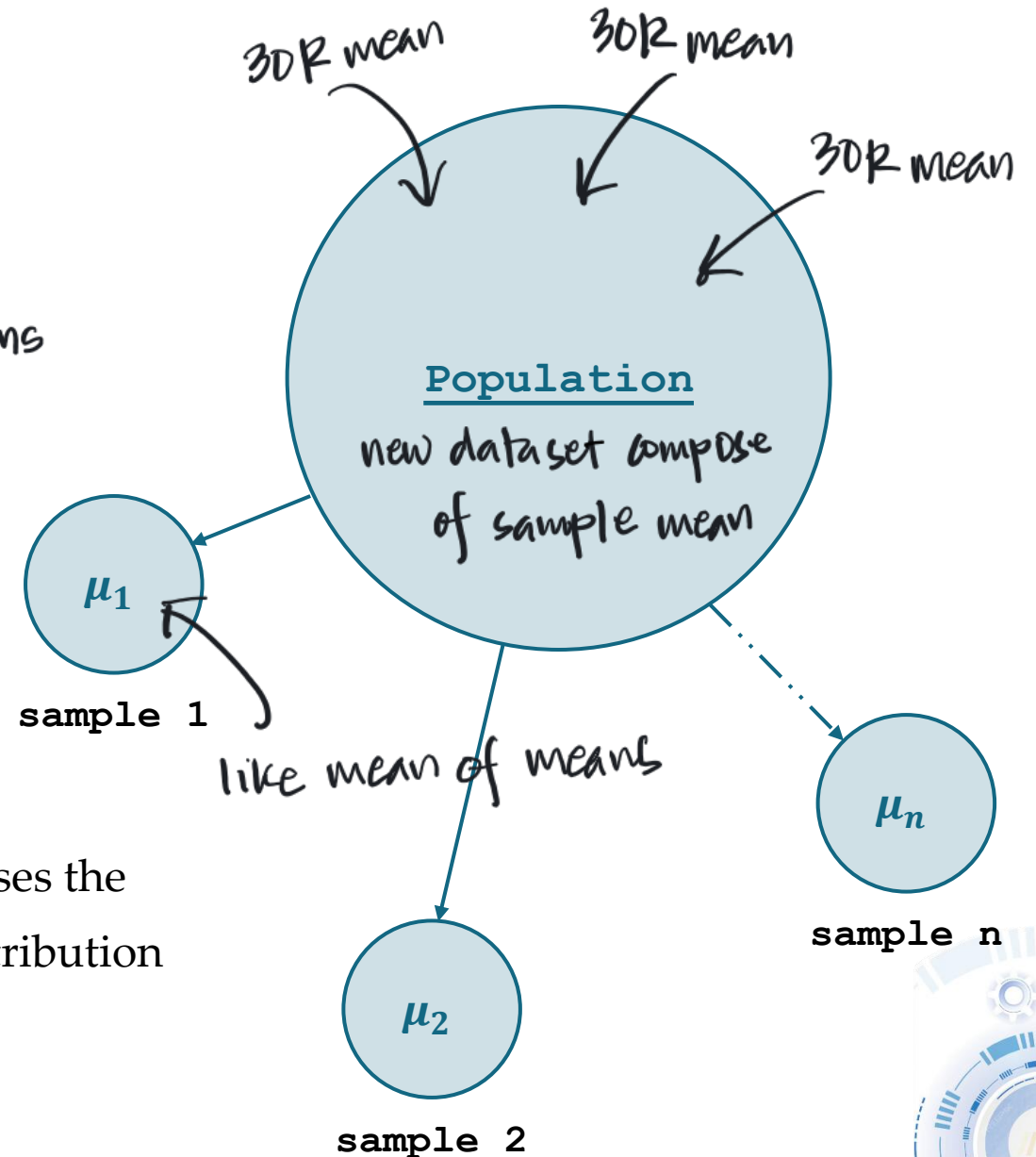
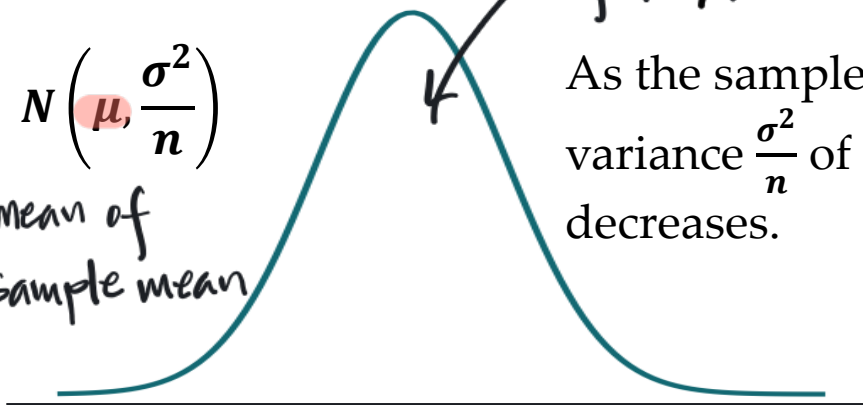
Sampling Distribution

$N\left(\mu, \frac{\sigma^2}{n}\right)$

mean of sample mean

distribution of $\bar{x}_1, \bar{x}_2 \dots \bar{x}_{100}$

As the sample size n increases the variance $\frac{\sigma^2}{n}$ of sampling distribution decreases.



SAMPLING DISTRIBUTION

Sampling Distribution

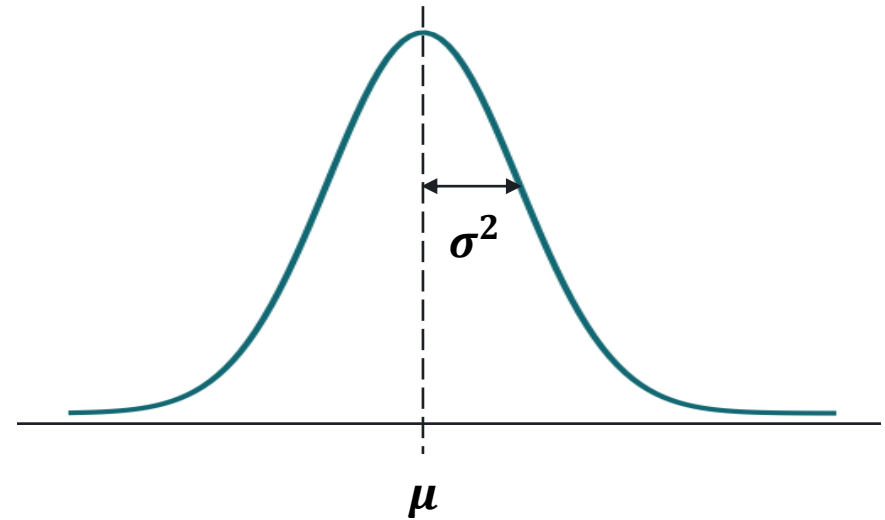
A sampling distribution is the probability distribution of a statistic (e.g., μ, σ^2) obtained from a large number of samples drawn from a specific population.

Denoted by

$$N\left(\mu, \frac{\sigma^2}{n}\right), n > 30$$

where

$\frac{\sigma^2}{n}$ = variance of the sampling distribution



STANDARD ERROR

Sampling Distribution

Standard error is the standard deviation of the distribution formed by the sample means.

Formula

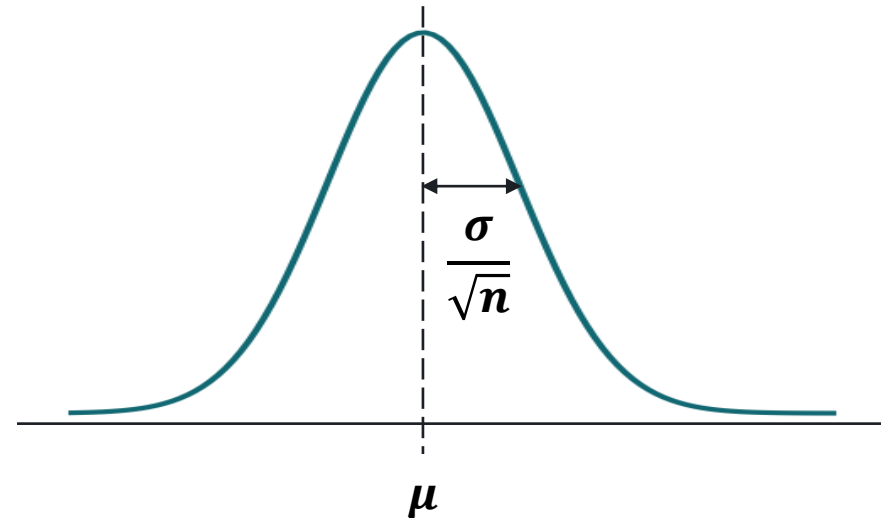
$$SE = \frac{\sigma}{\sqrt{n}}$$

$$\sqrt{\frac{\sigma^2}{n}}$$

where

σ = sampling standard deviation

n = number of observations



LABORATORY

